(System of linear equations)

Linear equations are expressed in the form of

$$a_1x_1 + a_2x_2 + \cdots a_nx_3 = b$$

x_1, x_2, x_3 → variables raised to power 1.

a_1, a_2 → constants.

b → constant.

$a_1x_1 + a_2x_2 + \cdots a_3x_3 = b$ where $b = 0$ then it is called homogenous linear eq.

e.g $x + 2y = 7$ ✓, $x_1 - 2x_2 + 3x_3 = 0$ ✓

$\sin x + y = 0$ ✗

System of linear equations:

$$5x + y = 3$$
$$2x - y = 4$$

> (x and y are unknowns and there Answers is the Sol to both)

$x = 1 \quad y = -2$

$5(1) - 2 = 3$ ✓

$2(1) - (-2) = 4$ ✓

→ can be written as $(1, -2)$ → order matter. $(1, -2) \neq (2, -1)$

System of L.E consistent → when it has one Sol.

unconsistent → NO Sol

Note: (No. of unknowns = No. of Eq)

Note:

$ax + by = c$ (2D) is a line equation.

$ax + by + cz = d$ (3D) is a plane equation.

Every system of linear equations has zero, one, infinitely many sol but no other possibilities exist

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using parametric eq to find linear equation.

1) $4x - 3y = 5$ Solve for leading var.

$4x = 5 + 3y$

$$x = \frac{5 + 3y}{4}$$

Before making the eq parametric make sure it is in its simplest form

2) Assign $y =$ aribtary parameter $y = t$

$$x = \frac{5}{4} + \frac{3}{4}t$$

$\rightarrow \left(\frac{5}{4} + \frac{3}{4}t, t \right) \rightarrow$ infinite Sol Set.

Augmented matrix $\Rightarrow$ using a matrix to Solve a System of linear eq.

$$x_1 + x_2 + 2x_3 = 8$$
$$-x_1 - 2x_2 + 3x_3 = 1$$
$$3x_1 - 7x_2 + 4x_3 = 10$$

$$\left[\begin{array}{ccc|c} x_1 & x_2 & x_3 & C \\ 1 & 1 & 2 & 8 \\ -1 & -2 & 3 & 1 \\ 3 & -7 & 4 & 10 \end{array}\right]$$

↳ augmented matrix.

At the end we get

$x_1 = a$

$x_2 = b$

$x_3 = c$

↳ At the end the As must look like this

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & a \\ 0 & 1 & 0 & b \\ 0 & 0 & 1 & c \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 3 & 4 & -2 & 1 \\ -5 & 6 & 2 & 7 \\ 4 & -3 & 0 & 2 \end{array}\right)$$

$2R_1 + R_3$ ↙ perform this and the answer will come at the place where this is written

$$\left(\begin{array}{ccc|c} 3 & 4 & -2 & 1 \\ -5 & 6 & 2 & 7 \\ 10 & 5 & -4 & 4 \end{array}\right)$$

if lets say the operation is written like

$3R_1 + R_2$

$$\left(\begin{array}{ccc|c} 3 & 4 & -2 & 1 \\ -5 & 6 & 2 & 7 \\ 4 & -3 & 0 & 2 \end{array}\right)$$

then the Answer will be written at the Last Row meaning (R_2)

Rules of elementary Row wise operations

1) Swap two rows $R_i \longleftrightarrow R_j$

2) Multiply a row by non-zero constant $R_i \to 5R_i$

3) Add two rows together $(R_i + R_j)$, or multiply by non-zero numbers. $\left(\frac{1}{2}R_1 + 11R_2\right)$

For a matrix to be in echelon form.

$$\begin{bmatrix} \textcircled{4} & -7 & 2 \\ 0 & \textcircled{1} & 6 \end{bmatrix} \qquad \begin{pmatrix} \textcircled{1} & 4 & 2 & 1 \\ 0 & \textcircled{-3} & -3 & 3 \\ 0 & 0 & \textcircled{-3} & -9 \end{pmatrix}$$

$$\begin{bmatrix} \textcircled{2} & 3 & -4 & 0 & 1 \\ 0 & 0 & \textcircled{1} & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

↓ No leading element.

★ Leading element : first element in any row that is non-zero

★ Anything below the leading element must be a zero.

★ Any row containing all zeros must be at the bottom of the Matrix.

Reduced echelon form → Satisfies all the previous conditions in echelon form

$$\left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & 6 \end{array}\right], \quad \left[\begin{array}{ccc|c} 1 & 0 & 0 & 11 \\ 0 & 1 & 0 & -4 \\ 0 & 0 & 1 & 3 \end{array}\right],$$

$$\left[\begin{array}{cccc|c} 1 & 1 & 0 & 0 & 5 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

new conditions

* each leading element is 1.

* no other non-zero entries in a column with leading 1.

(there must be all zero's)

Example:

$$x_1 + x_2 + 2x_3 = 8$$
$$-x_1 - 2x_2 + 3x_3 = 1$$
$$3x_1 - 7x_2 + 4x_3 = 10$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ -1 & -2 & 3 & 1 \\ 3 & -7 & 4 & 10 \end{array}\right]$$

$(R_1 + R_2)$

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & -1 & 5 & 9 \\ 3 & -7 & 4 & 10 \end{array}\right]$$

$(-3R_1 + R_3)$

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & -1 & 5 & 9 \\ 0 & -10 & -2 & -14 \end{array}\right]$$

$-R_2$

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & 1 & -5 & -9 \\ 0 & -10 & -2 & -14 \end{array}\right] \qquad \left(-\frac{1}{10}\right)$$

$(10R_2 + R_3)$

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & 1 & -5 & -9 \\ 0 & 0 & -52 & -104 \end{array}\right]$$

$R_3 \times \left(-\frac{1}{52}\right)$

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & 1 & -5 & -9 \\ 0 & 0 & 1 & 2 \end{array}\right] \rightarrow$$

echelon form

Reduced-echelon	echelon
$\begin{bmatrix} * & * & * \\ 0 & * & * \\ 0 & 0 & * \\ 0 & 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & * \\ 0 & 1 & 0 & * \\ 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 \end{bmatrix}$
* → Any non-zero no.	Zero's must be present in col where there is a leading one

(a)
$$\begin{array}{ccc|c} x & y & z & c \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{array}$$

$1x + 0y + 0z = 0$
$0x + 1y + 2z = 0$
$0x + 0y + 0z = 1$

$x = 0$
$y = -2z$
$0 = 1$ (x)

no sol. System is inconsistent

(b)
$$\begin{array}{ccc|c} x & y & z & c \\ 1 & 0 & 3 & -1 \\ 0 & 1 & -4 & 2 \\ 0 & 0 & 0 & 0 \end{array}$$

$x + 3z = -1$
$y - 4z = 2$
$0 = 0$

$x = -1 - 3z$
$y = 2 + 4z$
$z = t$

$x = -1 - 3t$
$y = 2 + 4t$
$z = t$

$(-1-3t, 2+4t, t)$ consistent

(gaussian elimination) → (echelon form)

$$x+y-z=-2$$
$$2x-y+z=5$$
$$-x+2y+2z=1$$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 1 & -1 & -2 \\ 2 & -1 & 1 & 5 \\ -1 & 2 & 2 & 1 \end{array}\right]$$

converting to row-echleon.

$R_1 + R_3$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & -2 \\ 2 & -1 & 1 & 5 \\ 0 & 3 & 1 & 3 \end{array}\right]$$

$-2R_1 + R_2$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & -3 & 3 & 9 \\ 0 & 3 & 1 & -1 \end{array}\right]$$

$R_3 + R_2$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & -3 & 3 & 9 \\ 0 & 0 & 4 & 8 \end{array}\right]$$

$-\frac{1}{3} R_2$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & 1 & -1 & -3 \\ 0 & 0 & 4 & 8 \end{array}\right]$$

$\frac{1}{4} R_3$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & 1 & -1 & -3 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

Steps

1) Make all zeros below the leading element.

2) then Back Subsitude

$x + y - z = 2$

$y - z = -3$

$z = 2$

$y = -3 + 2$

$y = -1$

$x = 1$

$(1, -1, 2)$

Doing with putting in row echelon form

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & 1 & -1 & -3 \\ 0 & 0 & 4 & 8 \end{array}\right] \rightarrow$$

$x + y - z = 2$

$y - z = -3$

$4z = 8$

$z = 2$

$x = 1, -1$

now use Substitution.

No need to Make zeros above

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gauss-Jordan elimination → Reduced echelon form

$x + y - z = 7$

$x - y + 2z = 3$

$2x + y + z = 9$

$\rightarrow$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 7 \\ 1 & -1 & 2 & 3 \\ 2 & 1 & 1 & 9 \end{array}\right]$$

(zeros)

3) No need of Substitution → gives us direct Answer

Homogenous linear Systems:- are always consistent.

$$5x + 4y = 0$$
$$2x - 2y = 0$$

$$x = 0, \quad y = 0$$

$Ax = 0$

Note: Homogenous System have always atleast one Sol that is the trivial Solution and in addition to Sometimes infinity that is non-trivial

if Both variables equal zero then it is trivial Sol.

if there are other Sol then it is called non-trivial Solutions.

consistent → Solvable
inconsistent → not Solvable

e.g using gauss-Jordan elimination Solve

$$x_1 + 3x_2 - 2x_3 + 2x_5 = 0$$
$$2x_1 + 6x_2 - 5x_3 - 2x_4 + 4x_5 - 3x_6 = 0$$
$$5x_3 + 10x_4 + 15x_6 = 0$$
$$2x_1 + 6x_2 + 8x_4 + 4x_5 + 18x_6 = 0$$

$$\left(\begin{array}{cccccc|c} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 2 & 6 & -5 & -2 & 4 & -3 & 0 \\ 0 & 0 & 5 & 10 & 0 & 15 & 0 \\ 2 & 6 & 0 & 8 & 4 & 18 & 0 \end{array}\right)$$

$x_1 \; x_2 \; x_3 \; x_4 \; x_5 \; x_6$

↳ all zeros hence homogenous system

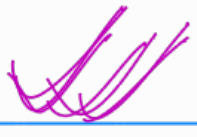
To Solve homogenous equation we use Reduced Row echelon form

free variables means infinity many Solutions

Even if homogenous system has free variables then one trivial Sol will always exist and other non-trivial.

$$\left(\begin{array}{cccccc|c} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 2 & 6 & -5 & -2 & 4 & -3 & 0 \\ 0 & 0 & 5 & 10 & 0 & 15 & 0 \\ 2 & 6 & 0 & 8 & 4 & 0 & 18 \end{array}\right) \begin{array}{l} R_1 \\ R_2 \\ R_3 \\ R_4 \end{array}$$

→ $R_2 - R_4$



gauss Jordan

→ first make the leading element 1.

→ then in its column make the rest no = zero.

→ move to next col

gauss elimination

→ find the leading element

→ in its column make sure there are zeros

→ Make zeros at the bottom then use Back-substitution

→ no need to make zeros at top

(Matrix operations) e.g of matrices $\begin{bmatrix} 1 & 0 \\ 3 & 2 \\ -1 & 3 \end{bmatrix}$, $[4]$, $\begin{pmatrix} \frac{1}{3} \end{pmatrix}$

Size of matrix R×C $\quad$ 3×2 $\quad$ 2×1

A matrix with 1 Row and 0 column is called Row matrix.

A matrix with 1 column and 0 Row is called column Matrix.

general notation:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \qquad A = \begin{bmatrix} 2 & -3 \\ 7 & 0 \end{bmatrix}$$

$(A)_{11} = 2 \qquad (A)_{12} = -3$

$(A)_{21} = 7$

(row, col)

A Square Matrix Row = columns

Two matrices are equal if they have same RxC and the elements are equal

$$A = \begin{bmatrix} 2 & 1 \\ 3 & x \end{bmatrix} \qquad B = \begin{bmatrix} 2 & 1 \\ 3 & 5 \end{bmatrix}$$

only for $x = 5$ $A = B$ else $A \neq B$

$$C = \begin{bmatrix} 2 & 1 & 0 \\ 3 & 4 & 0 \end{bmatrix} \quad A \neq C$$

if two matrices have same size then $A+B$, $A-B$

$$A = \begin{bmatrix} 2 & 1 & 0 & 3 \\ -1 & 0 & 2 & 4 \\ 4 & -2 & 7 & 0 \end{bmatrix} \qquad B = \begin{bmatrix} -4 & 3 & 5 & 1 \\ 2 & 2 & 0 & -1 \\ 3 & 2 & -4 & 5 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$$

A + C | A - C = undefined.

$$A + B = \begin{bmatrix} -2 & 4 & 5 & 4 \\ 1 & 2 & 2 & 3 \\ 7 & 0 & 3 & 5 \end{bmatrix}$$

Scalar multiple : multiplying a no with every element of Matrix.

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 1 & 3 & 1 \end{bmatrix} \qquad 2A = \begin{bmatrix} 4 & 6 & 8 \\ 2 & 6 & 2 \end{bmatrix}$$

Multiplication of matrix:

A : m x r B : r x n final Ans = m x n

Condition: no of col of first matrix = no of rows of second matrix.

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix} \rightarrow 2 \times 3$$

$$B = \begin{bmatrix} 4 & 1 & 4 & 3 \\ 0 & -1 & 3 & 1 \\ 2 & 7 & 5 & 2 \end{bmatrix} \rightarrow 3 \times 4$$

$2 \times 3 \longleftrightarrow 3 \times 4 \rightarrow \boxed{2 \times 4}$

$$\begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 \\ -1 \\ 7 \end{bmatrix}$$

2×3 $\quad$ 3×1

$2 \times 3 \longleftrightarrow 3 \times 1$

$\boxed{2 \times 1}$

$[1 \times 1 - 2 + 28] = 27$

$[2 \times 1 - 6 + 7] = -4$

$$\begin{bmatrix} 27 \\ -4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 4 \end{bmatrix} \begin{bmatrix} 4 & 1 & 4 & 3 \\ 0 & -1 & 3 & 1 \\ 2 & 7 & 5 & 2 \end{bmatrix}$$

(1×3) (3×4) (1×4)

Ans = 1×4

$$[1\times4+0+8],\ [1-2+28],\ [4+6+20],\ [3+2+8]$$

$$[12,\ 27,\ 30,\ 13]$$

Note:: $A = m\times n$ $\quad x = n\times1$ column vector | has only one column

$$\begin{bmatrix} -1 & 3 & 2 \\ 1 & 2 & -3 \\ 2 & 1 & -2 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$$

A B

(3×3) $\quad (3\times1) \rightarrow (3\times1)$

$$2\begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} - 1\begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} + 3\begin{bmatrix} 2 \\ -3 \\ -2 \end{bmatrix} = \begin{bmatrix} 1 \\ -9 \\ -3 \end{bmatrix}$$

Final Answer will be combination of columns of A times number in x and so on.

Note:: There are different methods for multiplying matrix.

Matrix form of a linear system::

e.g $\quad 2x_1 + 3x_2 = 5$

$\quad 4x_1 - x_2 = 1$

> Rewriting the linear system in matrix form $AX = B$

$A = \begin{bmatrix} 2 & 3 \\ 4 & -1 \end{bmatrix}$ $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ $B = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$

A = coefficients
X = unknown variable
B = constant

Transpose of a Matrix

Transpose of matrix A → A^T (Switch rows with columns.)

$C = \begin{bmatrix} 1 & 3 & 5 \end{bmatrix}$ $C^T = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$

$B = \begin{bmatrix} 2 & 3 \\ 1 & 4 \\ 5 & 6 \end{bmatrix}$ $B^T = \begin{bmatrix} 2 & 1 & 5 \\ 3 & 4 & 6 \end{bmatrix}$

Special case: Square matrix n×n

$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$

$A^T = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$

Flip the things above and below the main diagonal.

Trace of a matrix (only for n×n)

if we have a Square matrix then the trace is adding the sum of main diagonal

$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ Trace = 1+5+9 = 15

Inverses:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix} \Rightarrow AB$$

$$\begin{array}{c} \\ (1,2) \\ (3,4) \\ (0,1) \end{array} \begin{array}{c} \begin{array}{cc} (4,2) & (3,1) \end{array} \\ \begin{bmatrix} 8 & 5 \\ 20 & 13 \\ 2 & 1 \end{bmatrix} \end{array}$$

Shortcut of matrix multiplication.

order matters in matrix multiplication.

$$A = \begin{bmatrix} -1 & 0 \\ 2 & 3 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}$$

$$AB = \begin{array}{c} \\ (-1,0) \\ (2,3) \end{array} \begin{array}{c} \begin{array}{cc} (1,3) & (2,0) \end{array} \\ \begin{bmatrix} -1 & -2 \\ 11 & 4 \end{bmatrix} \end{array}$$

$$BA = \begin{array}{c} \\ (1,2) \\ (3,0) \end{array} \begin{array}{c} \begin{array}{cc} (-1,2) & (0,3) \end{array} \\ \begin{bmatrix} 3 & 6 \\ -3 & 0 \end{bmatrix} \end{array}$$

$AB \neq BA$

hence matrix multiplication is not commutative

Zero Matrices: → all entries are zero.

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ e.g}$$

$A + 0 = 0 + A = A$

A matrix. zero matrix (only if they have same size).

we cannot cancel matrices.

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix} \quad C = \begin{pmatrix} 2 & 5 \\ 3 & 4 \end{pmatrix}$$

$AB = AC$

$B = C$ (✗) we cannot cancel a matrix because we will lose info.

Identity matrices:. A Square matrix with 1's on main diagonal and zeros everywhere else.

(1) (1x1), $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (2x2), $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (3x3)

Denoted by I.

Rule

$$A = \begin{bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{bmatrix} \qquad I = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix} \qquad A \cdot I_2 = A$$

$$I = \begin{bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{bmatrix} \qquad A \cdot I_3 = A$$

$$A \cdot I = A$$

A and I must have Same Size

Multiplying a matrix A with the Same no of rows Identity or Same no of columns Identity give the Same Matrix again.

imp.

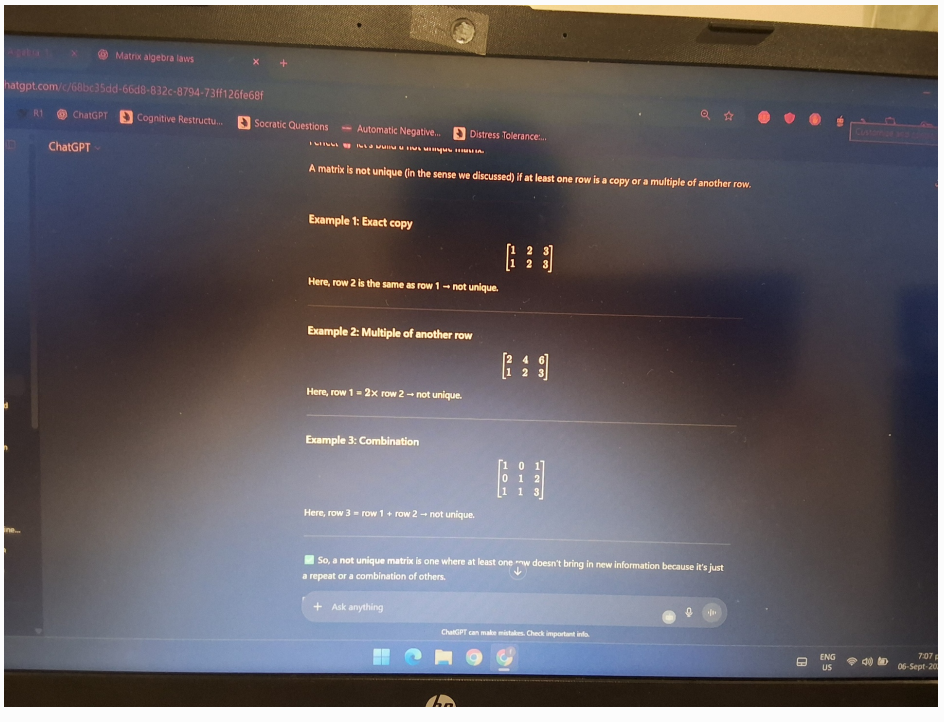
Using reduced row echelon form of a square matrix leads to either a identity (unique) or a Matrix containing a complete 0 row

unique means no element was replaced in matrix.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad R = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad R = \begin{bmatrix} 2 & 3 \\ 0 & 0 \end{bmatrix}$$

→ RREF



Inverse of a Matrix:

$$AB = BA = I$$

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \qquad B = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$$

$$AB = \begin{matrix} \\ 1,0 \\ 0,2 \end{matrix} \begin{matrix} \begin{matrix} (1,0) & (0,\frac{1}{2}) \end{matrix} \\ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \end{matrix}$$

$$BA = \begin{matrix} \\ 1,0 \\ 0,\frac{1}{2} \end{matrix} \begin{matrix} \begin{matrix} (1,0) & (0,2) \end{matrix} \\ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \end{matrix}$$

If we multiply A with another matrix B and both result the same identity matrix, then A is said to be (non-singular) → inverse exist and B is the inverse of A. and A is inverse of B and B is also non-singular because it is inverse of A. **A and B both are inverse of each other.**

For inverse to exist both AB and BA must yield I.

Properties of Inverses

if B and C are both inverses of the matrix A then.

$$B = C$$

$$AB = BA = I$$
$$AC = CA = I$$

$$B = B \cdot I$$
$$B = B(AC)$$
$$(AB)C = B(AC)$$ using Associativity.
$$I \cdot C = B(I)$$
$$B = C$$ proven

$A \cdot A^{-1} = I$

Matrix A — Inversed matrix A^{-1}

Formula for Inverse

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Determinant

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$$

if determinant $= 0$. No inverse

determinant $\neq 0$ inverse

Q check where $A = \begin{bmatrix} -1 & 2 \\ 3 & -6 \end{bmatrix}$ is invertable and find its inverse

$-1(-6) - 2 \times 3 \rightarrow 6 - 6 = 0$ no inverse exist.

Suppose we have two matrices A and B.

Both are invertable $A \cdot A^{-1} = I$

$B \cdot B^{-1} = I$

$(AB)^{-1} = B^{-1}A^{-1}$ (Inverse of product)

e.g

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} \qquad B = \begin{bmatrix} 3 & 2 \\ 2 & 2 \end{bmatrix}$$

$A = 3 - 2 = 1$ ✓ $\qquad B = 6 - 4 = 2$ ✓

$$AB = \begin{matrix} (1,2) \\ (1,3) \end{matrix} \begin{matrix} (3,2) & (2,2) \end{matrix} \begin{bmatrix} 7 & 6 \\ 9 & 8 \end{bmatrix} = (AB)^{-1}$$

$$\frac{1}{(7\times 8)-(9\times 6)}\begin{bmatrix} 8 & -6 \\ -9 & 7 \end{bmatrix} \rightarrow \begin{bmatrix} 4 & -3 \\ -9/2 & 7/2 \end{bmatrix} = (AB)^{-1}$$

$$A^{-1} = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix} \quad B^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & \frac{3}{2} \end{bmatrix}$$

$$B^{-1}A^{-1} = \begin{bmatrix} 4 & -3 \\ -9/2 & 7/2 \end{bmatrix}$$

Power of Matrix:. $A \rightarrow$ Square matrix.

$A^0 = I \qquad A^1 = A \qquad A^2 = AA.$

$A^{-n} \rightarrow$ only possible if A is invertable.

$A^{-2} = A^{-1} \cdot A^{-1}$

$A^{-3} = A^{-1} \cdot A^{-1} \cdot A^{-1}$

Properties

$(A^{-1})^{-1} = A$

$(A^n)^{-1} = (A^{-1})^n$

$(kA)^{-1} = (A^{-1})\frac{1}{k}$

(Matrix polynomial:)

Find $P(A)$ For $p(x) = x^2 - 2x - 5 \qquad A = \begin{bmatrix} -1 & 2 \\ 1 & 3 \end{bmatrix}$

$P(A) = A^2 - 2A - 5I$ → we always multiply a scala with identity to make them a Matrix.

$$\begin{bmatrix} -1 & 2 \\ 1 & 3 \end{bmatrix}\begin{bmatrix} -1 & 2 \\ 1 & 3 \end{bmatrix} - \begin{bmatrix} -2 & 4 \\ 2 & 6 \end{bmatrix} - 5\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

= final As

Inverse of Transpose:: If A is invertable then its transpose is also invertable

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Elementary matrices

if we did elementary operations on A matrix to go to B matrix. we can do the reverse and go Back to matrix A from B.

Row-equivalent matrices → If we go from matrix A to matrix B using Row op.

Elementary matrices: = Starting with a identity matrix and performing a Singe Row op. then multiply A with Elemntary matrix to get B.

$$A = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 2 & -1 & 3 & 6 \\ 1 & 4 & 4 & 0 \end{bmatrix} \qquad E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}$$

$$EA = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 2 & -1 & 3 & 6 \\ 1 & 4 & 4 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}$$

E was made by $3R_1 + R_3$

Multiplying E · A should do the same op on A.

Always multiply elementary matrix first (to the left) and then matrix A.

$$EA = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 & 3 \\ 2 & -1 & 3 & 6 \\ 1 & 4 & 4 & 0 \end{bmatrix}$$

$$\begin{matrix} 1,0,0 \\ 0,1,0 \\ 3,0,1 \end{matrix} \begin{bmatrix} (1,2,1) & (0,-1,4) & (2,3,4) & (3,6,0) \\ 1 & 0 & 2 & 3 \\ 2 & -1 & 3 & 6 \\ 4 & 4 & 10 & 9 \end{bmatrix}$$

Elementary matrices are made by performing one row operation on I. and every row operation can by reversed by another row operation.

Hence elementary matrix has an inverse elementary matrix that turns it back to I.

E is always invertable.

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad 7R_2 \rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 7 \end{bmatrix} \rightarrow R_2/7$$

$$\rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad 5R_2 \rightarrow E_1 = \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix}$$

$$\tfrac{1}{5}R_2 \rightarrow E_0 = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{5} \end{bmatrix}$$

$$\boxed{E_0 \cdot E_1 = I}$$

Equivalence theorem:

if a is $n \times n$ matrix then the following statements are all true or all false.

1) a is invertable
2) $aX = 0$ has only the trivial sol
3) RREF of a is I
4) a can be expressed as a product of elementary matrices.

$$a = E_1 E_2 \cdots E_k$$

if any one of the statements is true all are true and vice versa.

A method of inverting matrices

we can use the formula but it only works for 2×2. For matrices with greater size it becomes complicated hence we use this method

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

write with identity.

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 4 & 0 & 1 \end{array}\right]$$

Make the A RREF

$3R_1 - R_2$

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 2 & 3 & -1 \end{array}\right]$$

$\frac{R_2}{2}$

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array}\right]$$

$2R_2 - R_1$

$$\left[\begin{array}{cc|cc} -1 & 0 & 2 & -1 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array}\right]$$

$-1R_1$

$$\left[\begin{array}{cc|cc} 1 & 0 & -2 & 1 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array}\right]$$

$$A^{-1} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}$$ now the identity has become inverse of A.

---

when a matrix is not invertable.

$$A = \begin{bmatrix} 1 & 6 & 4 \\ 2 & 4 & -1 \\ -1 & 2 & 5 \end{bmatrix} \qquad \left[\begin{array}{ccc|ccc} 1 & 6 & 4 & 1 & 0 & 0 \\ 2 & 4 & -1 & 0 & 1 & 0 \\ -1 & 2 & 5 & 0 & 0 & 1 \end{array}\right]$$

$R_1 + R_3$

$$\left[\begin{array}{ccc|ccc} 1 & 6 & 4 & 1 & 0 & 0 \\ 2 & 4 & -1 & 0 & 1 & 0 \\ 0 & 8 & 9 & 1 & 0 & 1 \end{array}\right]$$

$-2R_1 + R_2$

$$\left[\begin{array}{ccc|ccc} 1 & 6 & 4 & 1 & 0 & 0 \\ 0 & -8 & -9 & 0 & 1 & 0 \\ 0 & 8 & 9 & 1 & 0 & 1 \end{array}\right]$$

$R_2 / -8$

$$\left[\begin{array}{ccc|ccc} 1 & 6 & 4 & 1 & 0 & 0 \\ 0 & 1 & \frac{9}{8} & 0 & -\frac{1}{8} & 0 \\ 0 & 8 & 9 & 1 & 0 & 1 \end{array}\right]$$

$\frac{1}{8}R_2 - R_3 \to R_3$

$$\left[\begin{array}{ccc|ccc} 1 & 6 & 4 & 1 & 0 & 0 \\ 0 & 1 & 9/8 & 0 & -1/8 & 0 \\ 0 & 0 & 0 & -1 & -1 & -1 \end{array}\right]$$

Hence inverse of A doesn't exist.

Example 6 Ex 1.5

no of pivots (or) :- no of variable

↳ that means matrix is invertable

(a)

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 2 & 5 & 3 & 0 \\ 1 & 0 & 8 & 0 \end{array}\right]$$

$R_2 \leftarrow 2R_1 - R_2$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & -1 & 3 & 0 \\ 1 & 0 & 8 & 0 \end{array}\right]$$

$R_3 \leftarrow R_1 - R_3$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & -1 & 3 & 0 \\ 0 & 2 & 5 & 0 \end{array}\right]$$

$R_2 \times (-1)$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 2 & 5 & 0 \end{array}\right]$$

$R_3 \leftarrow R_3 - R_1$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$R_1 \leftarrow 2R_2 - R_1$

$$\left[\begin{array}{ccc|c} -1 & 0 & -9 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 2 & 0 \end{array}\right]$$

$R_1 \times (-1)$

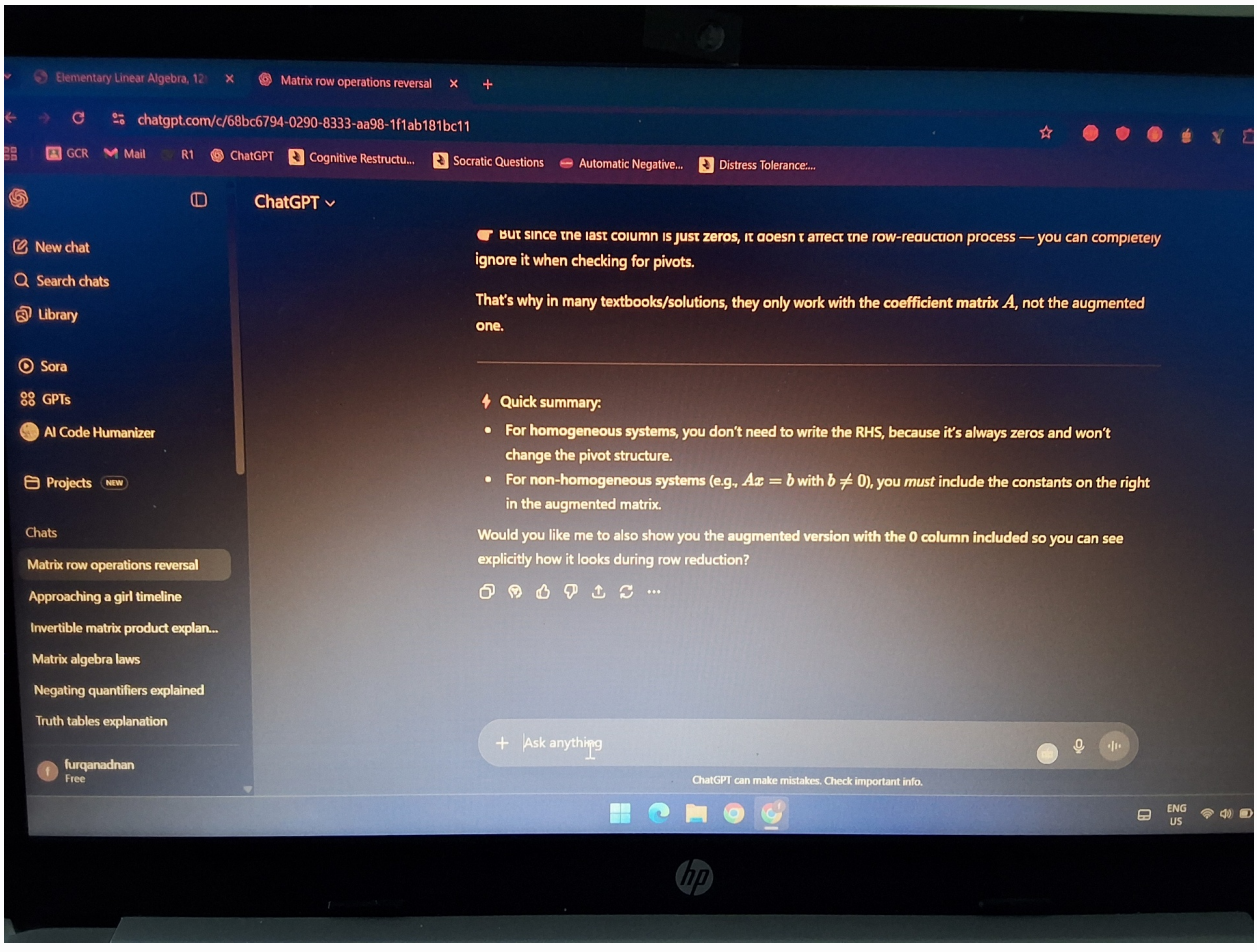
$$\left[\begin{array}{ccc|c} 1 & 0 & 9 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 2 & 0 \end{array}\right]$$

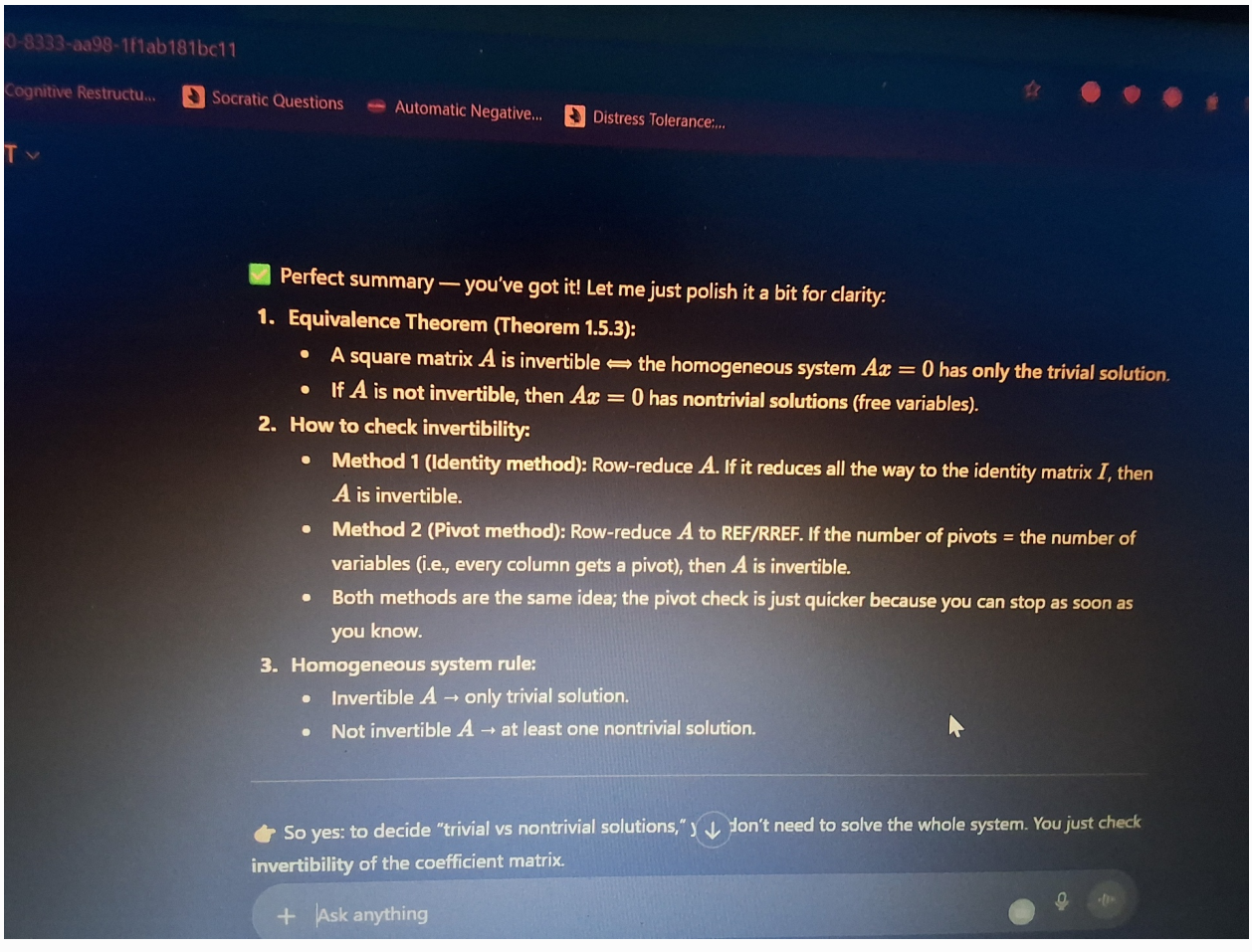
$\frac{1}{2}R_3$ $$\left[\begin{array}{ccc} 1 & 0 & 9 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{array}\right]$$

$3R_3 + R_2$ $$\left[\begin{array}{ccc} 1 & 0 & 9 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right]$$

$-9R_3 + R_1$

$$\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right] \rightarrow \text{three pivots} = 3 \text{ variables.}$$


Elementary Linear Algebra, 12
Matrix row operations reversal
chatgpt.com/c/68bc6794-0290-8333-aa98-1f1ab181bc11
GCR
Mail
R1
ChatGPT
Cognitive Restructu...
Socratic Questions
Automatic Negative...
Distress Tolerance:...
ChatGPT
New chat
Search chats
Library
Sora
GPTs
AI Code Humanizer
Projects
NEW
Chats
Matrix row operations reversal
Approaching a girl timeline
Invertible matrix product explan...
Matrix algebra laws
Negating quantifiers explained
Truth tables explanation
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Free
But since the last column is just zeros, it doesn't affect the row-reduction process — you can completely ignore it when checking for pivots.
That's why in many textbooks/solutions, they only work with the coefficient matrix A, not the augmented one.
Quick summary:
For homogeneous systems, you don't need to write the RHS, because it's always zeros and won't change the pivot structure.
For non-homogeneous systems (e.g., $Ax = b$ with $b \neq 0$), you must include the constants on the right in the augmented matrix.
Would you like me to also show you the augmented version with the 0 column included so you can see explicitly how it looks during row reduction?
Ask anything
ChatGPT can make mistakes. Check important info.
ENG
US


0-8333-aa98-1f1ab181bc11
Cognitive Restructu...
Socratic Questions
Automatic Negative...
Distress Tolerance:...
T
✅ Perfect summary — you've got it! Let me just polish it a bit for clarity:
1. Equivalence Theorem (Theorem 1.5.3):
A square matrix A is invertible ⟺ the homogeneous system $Ax = 0$ has only the trivial solution.
If A is not invertible, then $Ax = 0$ has nontrivial solutions (free variables).
2. How to check invertibility:
Method 1 (Identity method): Row-reduce A. If it reduces all the way to the identity matrix I, then A is invertible.
Method 2 (Pivot method): Row-reduce A to REF/RREF. If the number of pivots = the number of variables (i.e., every column gets a pivot), then A is invertible.
Both methods are the same idea; the pivot check is just quicker because you can stop as soon as you know.
3. Homogeneous system rule:
Invertible A → only trivial solution.
Not invertible A → at least one nontrivial solution.
👉 So yes: to decide "trivial vs nontrivial solutions," you don't need to solve the whole system. You just check invertibility of the coefficient matrix.
Ask anything

$Ax = B$ if Matrix A is invertable then the System will always have One Sol only

$x = A^{-1}B$ one we can find it without using RREF | REF

e.g $2x + y = 5$
$x + 3y = 7$

$a = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$ $x = x, y$

$b = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$ $3(2) - 1 \neq 0$

finding inverse of a

$$\frac{1}{6-1}\begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix} \rightarrow \frac{1}{5}\begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}$$

$$x = \frac{1}{5}\begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

Always multiply first then scale down

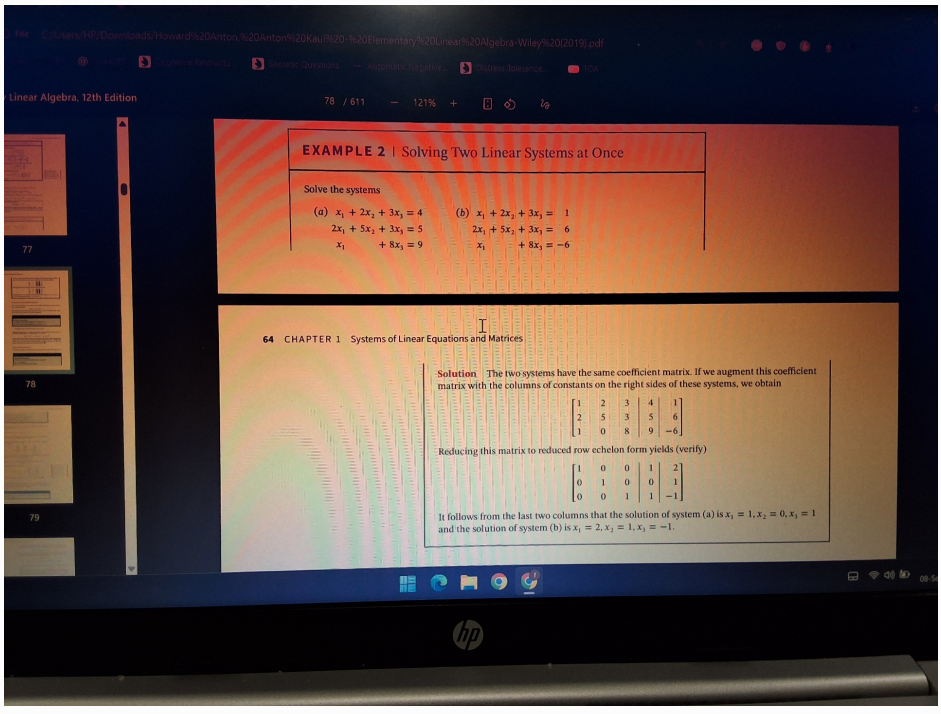
$(5, 7)$

$x = \begin{matrix} (3,-1) \\ (-1,2) \end{matrix}\begin{bmatrix} 8 \\ 9 \end{bmatrix}$ now Scale by $\frac{1}{5}$

Solution

$x = \begin{pmatrix} 8/5 \\ 9/5 \end{pmatrix}$

$$\boxed{x = \frac{8}{5}, \frac{9}{5}}$$



Linear Algebra, 12th Edition

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EXAMPLE 2 | Solving Two Linear Systems at Once

Solve the systems

$$\begin{array}{rrrrrrr} (a) & x_1 & + 2x_2 & + 3x_3 & = 4 & \qquad (b) & x_1 + 2x_2 + 3x_3 = 1 \\ & 2x_1 & + 5x_2 & + 3x_3 & = 5 & & 2x_1 + 5x_2 + 3x_3 = 6 \\ & x_1 & & + 8x_3 & = 9 & & x_1 \quad + 8x_3 = -6 \end{array}$$

64 CHAPTER 1 Systems of Linear Equations and Matrices

Solution The two systems have the same coefficient matrix. If we augment this coefficient matrix with the columns of constants on the right sides of these systems, we obtain

$$\left[\begin{array}{ccc|cc} 1 & 2 & 3 & 4 & 1 \\ 2 & 5 & 3 & 5 & 6 \\ 1 & 0 & 8 & 9 & -6 \end{array}\right]$$

Reducing this matrix to reduced row echelon form yields (verify)

$$\left[\begin{array}{ccc|cc} 1 & 0 & 0 & 1 & 2 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & -1 \end{array}\right]$$

It follows from the last two columns that the solution of system (a) is $x_1 = 1, x_2 = 0, x_3 = 1$ and the solution of system (b) is $x_1 = 2, x_2 = 1, x_3 = -1$.

only if two matrices have same coefficients (Regardless of RHS) we can solve them simultaneously either using RREF / REF

Properties of invertable matrices

we know $AB = BA = I$. A cancels B and B cancels A. Both are inverse of each other.

note:- Determining whether a system is consistent through elimination

if any row becomes all zeros then one sol exist else no solution exist

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & b_1 \\ 2 & 5 & 3 & b_2 \\ 1 & 0 & 8 & b_3 \end{array}\right] \xrightarrow{R} \left[\begin{array}{ccc|c} 1 & 2 & 3 & b_1 \\ 2 & 5 & 3 & b_2 \\ 0 & 0 & 0 & b_3 \end{array}\right]$$

$$x_1 + x_2 + x_3 = b_3$$

$[0\ 0\ 0\ |\ 0]$ → Infinite sol

$[0\ 0\ 0\ |\ c]$ $c \neq 0$

Determinants

$|A| = ad - bc$ for (2×2)

$|A| \neq 0$ then invertable

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Finding determinant for $n \times n$ then what ?

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

use row-echelon form and a_{33} will be determinant of that matrix

$$A^{-1} = \frac{\text{adjoint}(A)}{\det(A)}$$

Finding minors :

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

For lets say a_{11}

minor of a_{11} = $\begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix}$

determinant of Sub matrix a_{11} = $M_{11} = (a_{33}a_{22} - a_{23}a_{32})$

co-factor formula $C_{ij} = (-1)^{i+j} M_{ij}$

$$\begin{bmatrix} 3 & 1 & -4 \\ 2 & 5 & 6 \\ 1 & 4 & 8 \end{bmatrix}$$

$$M_{11} = \begin{bmatrix} 5 & 6 \\ 4 & 8 \end{bmatrix} = 5\times8 - 6\times4 = 16$$

$$C_{11} = (-1)^{1+1}(16) = \boxed{16}$$

$$\text{cofactor matrix} = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}$$

now determinant $(A) = a_{11}C_{11} + a_{12}C_{12}$

this is for (2×2)

determinant $(A) = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$

this is for (3×3)

and so on.

eg $A = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 3 & 1 & 2 & 2 \\ 1 & 0 & -2 & 1 \\ 2 & 0 & 0 & 1 \end{bmatrix}$

find determinant.

$\det(A) = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} + a_{14}C_{14}$

$\det(A) = 1(C_{11}) + 0 + 0 - 1(C_{14})$ → $C_{11} - C_{14}$ (determinant)

C_{11} $\begin{bmatrix} 1 & 2 & 2 \\ 0 & -2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

C_{14} $\begin{bmatrix} 3 & 1 & 2 \\ 1 & 0 & -2 \\ 2 & 0 & 0 \end{bmatrix}$

finding cofactor of 3×3

$C_{11} = (-1)^{1+1}(-2)\begin{vmatrix} -2 & 1 \\ 0 & 1 \end{vmatrix}$

$= \textcircled{-2}$

$C_{12} = (-1)^{3}(0)\begin{vmatrix} 0 & 1 \\ 0 & 1 \end{vmatrix}$

$= \textcircled{0}$

$C_{13} = (-1)^{4}(0)\begin{vmatrix} 0 & -2 \\ 0 & 0 \end{vmatrix}$

again using $a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$

$C_{11} = (-1)^{1+1}(0)\begin{vmatrix} 0 & -2 \\ 0 & 0 \end{vmatrix}$

$C_{11} = 0$

$C_{12} = (-1)^{3}(4)\begin{vmatrix} 1 & -2 \\ 2 & 0 \end{vmatrix}$

$C_{12} = -4$

$\begin{vmatrix} 1 & 0 \\ 2 & 0 \end{vmatrix}$

$C_{13} = 0$

det of this 3×3 $= (1\times2) + (2\times0) + (2\times0)$

$= -2$

$C_{13} = (-1)^{1+3}(0)$

$C_{13} = 0$

det of this 3×3 $= 3(-) - 4 + 0$

$= -4$

determinant of 4×4 $= -2 - (-4) = -6$

(calculating inverse) $A^{-1} = \frac{\text{Adjoint}(A)}{\text{determinant}(A)}$

$Adj = (C_{ij})^T$ transpose of cofactor matrix.

Column Row multiplication

$AB = ($first column of A × first Row of B$)$

$$A = \begin{bmatrix} 1 & 2 & 6 \\ 4 & 3 & 7 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 2 \\ 3 & 7 \\ 5 & 8 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 4 \end{bmatrix}\begin{bmatrix} 1 & 2 \end{bmatrix} + \begin{bmatrix} 2 \\ 3 \end{bmatrix}\begin{bmatrix} 3 & 7 \end{bmatrix} + \begin{bmatrix} 6 \\ 7 \end{bmatrix}\begin{bmatrix} 5 & 8 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 3 & 1 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 2 & -2 & 0 \\ -1 & 2 & 1 & 1 \end{bmatrix}$$

$$C_{ij} = (-1)^{i+j} M_{ij}$$

$$\det = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} + a_{14}C_{14}$$

$$1C_{11} + 3C_{12} + 1C_{13} + 2C_{14}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & -2 & 0 \\ 2 & 1 & 1 \end{bmatrix} \qquad \begin{bmatrix} 0 & 0 & 0 \\ 0 & -2 & 0 \\ -1 & 1 & 1 \end{bmatrix} \qquad \begin{bmatrix} 0 & 1 & 0 \\ 0 & 2 & 0 \\ -1 & 2 & 1 \end{bmatrix}$$

$$\begin{pmatrix} -2 & 0 \\ 1 & 1 \end{pmatrix}$$

det = 0 det = 0

$$M_{11} = -2$$

$$C_{11} = (-1)^2 (-2)$$

$$= 1(-2) = -2$$

det = −2

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 2 & -2 \\ -1 & 2 & 1 \end{bmatrix}$$

$$(-1)^3 (-2) \rightarrow (2)$$

$$\begin{bmatrix} 0 & -2 \\ -1 & 1 \end{bmatrix} \rightarrow -2$$

$$1C_{11} + 3C_{12} + 1C_{13} + 2C_{14}$$

$$1(-2) + 3(0) + 1(0) + 2(2)$$

$$-4 - 4 = -8$$